Literal structural descriptions in ten retained PGL_2 tables

Private recovery component

Theorem 1. *For the ten current companion tables and the accepted numerical row bijections below, all 329 displayed structural descriptions hold for the actual retained subgroup representatives. This includes the ten top rows. Each extension description asserts the displayed decomposition, rather than uniqueness of an isomorphism type from its factor names.*

q	7	9	11	13	17
rows	23	26	29	30	32
q	19	23	25	27	31
rows	36	34	43	32	44

The current PDFs are attributed to Thomas Connor and Dimitri Leemans. All eighteen native pages, including the empty continued table on the second page for $q = 11$, have been inspected. We do not identify these bytes with every historical file accompanying the 2013 article. The numerical classification, class lengths and weighted incidences are acquired from the previously accepted literal inventories and robust weight calculations. This component concerns their structural labels. The source's word "lattice" supplies no lattice-property premise.

Exact native assertions. The row ledger retains parentheses, subscripts, superscripts, centered dots and colons. Bare integers denote cyclic groups, p^e denotes an elementary abelian group, and the dihedral subscript denotes group order. Extension strings are associated from the left, as in the acquired naming interface. A colon is certified as a nontrivial split extension; a centered dot is certified as a nonsplit extension. In particular, row 14 for $q = 23$ is cyclic of order 22, while rows 28 and 29 print 2^2. Row 15 for $q = 31$ is cyclic of order 32, while the 3^2 entries for $q = 9, 27$ are elementary abelian. For $q = 9$, row 3 is $(3 : S_3) \cdot 2 \cdot 2$, with two nonsplit steps. None of these distinctions is inferred from a flattened decimal string or an expected search result.

Acquired actual models. Every source row is joined by its accepted row bijection to its actual saved permutation generators. A strict parser requires literal embedded generators for every subgroup, including proper projective subgroups; display names are not evaluated as constructors. The ambient models are the exact originally executed $\mathrm{PGL}(2, q)$ models. Their named-constructor identification and complete subgroup inventories are explicit acquired inputs. Order alone does not identify a group. The new top certificates prove the split descriptions, including $A_6 : 2$ at $q = 9$.

Shared registry and literal transports. The accepted PSL_2 structural component and its predecessor A5/M11 components supply the actual small models and their decompositions. The accepted literal alternating-group model supplies A_6; nine accepted PSL_2 ambient models supply the other simple factors. The fixed roster uses 76 such model inputs, retaining multiple models of a description when available. Each input is checked against its exact acquired bytes. The present certificate adds 43 concrete models, whose factor references only point backwards to acquired or earlier verified models.

We use the acquired generator-edge theorem MC.3. With right permutation composition, close the ordered generators under positive words. A stored map sends identity to identity, is bijective onto the closure of the target generators, and satisfies $f(xa) = f(x)f(a)$ for every generator a and every model element x. Induction on words proves that f is the unique homomorphism with these generator images; bijectivity makes it an isomorphism. This checks a number of products proportional to the order times the generator count, not a full quadratic multiplication table. The literal verifier checks all maps, including the simple factors inside the actual top groups. No transient GAP isomorphism assertion is acquired without these explicit checks.

The new decomposition certificates. The atom and extension criteria are exactly those proved in the accepted PSL_2 component. A cyclic certificate is an element of full order. A dihedral certificate consists of r, s of orders $m, 2$, with $srs = r^{-1}$ and complete generated group of order $2m$. Previously proved elementary and alternating atoms are transported from the registry.

For a split node $H = N : K$, the verifier constructs N, K from their literal generators, checks membership, and verifies their previously certified descriptions by generator-edge maps. Conjugating each generator of N by each generator of H stays in N; finiteness proves $N \trianglelefteq H$. The checks $N \cap K = 1$ and $|N||K| = |H|$ give $H = NK$ and the claimed semidirect product. A noncommuting generator pair verifies nontrivial action. The one new complement of composite order is the literal S_3 complement at row 6 for $q = 27$, in $3^2 : S_3$; it receives the same full checks.

Every new centered-dot step has index two. For a normal subgroup N of prime index p, a complement exists exactly when an outside element has order p: such an element generates a complement, and a complement provides such an element. The complete check $x^p \neq 1$ on every $x \in H \setminus N$ therefore proves nonsplitting. This is the acquired prime-index criterion, not a conclusion from a failed complement search. All 640 outside elements in the new dotted-node certificates are tested.

Finite coverage and development record. The final literal verifier checks all 329 row occurrences and the factors of all 43 new models: 374 transports and 424,622 right-generator edges. All ten ambient rows receive actual split-extension certificates, rather than merely retaining a printed top name. The original embedded arrays, the numerical row maps, the native assertion trees and their exact orders are checked before these certificates are used.

Witness generation was confined to the saved groups. The first bounded pass took 5.79 seconds and found 327 row witnesses. Its source-preparer exception for cyclic 22 in the $q = 23$ table had been applied too broadly to the two elementary abelian rows. The native reading and original outputs are preserved; the corrected exception is explicitly row 14. A separate 1.57-second pass supplied only the two missing witnesses. The other 327 witnesses and all 43 new model certificates were reused. The incomplete first build failed its full-roster guard, and no partial result was admitted. There was no ambient subgroup census or repeated classification. Candidate normal-subgroup, complement and isomorphism routines only suggest data that the literal verifier independently checks.

The damaged controls change identity images, bijections, element and row coverage, actual row and generator joins, the cyclic/power and dot/colon distinctions, an acquired seed, a quotient, dihedral relations and the factor dependency order. Each damaged input must be rejected. Separate cyclic versus elementary controls exercise the prime-two and prime-three non-splitting criterion inherited from the accepted verifier. Ordinary and optimized Python agree, with explicit checks rather than removable assertions.

Scope and attribution. The finite claim is confined to the displayed structural descriptions in these ten exact current tables, conditional on the accepted complete numerical inventories and named model inputs. It is not a new subgroup classification or a unique group-naming theorem, does not identify unseen historical editions, and does not complete the full atlas. It supplies no new original-source, question, formalization or measurement credit. The registry, normal-series and generator-edge arguments are acquired standard methods. Root proposed continued registry reuse; this component's author prepared the native ledger, new witnesses and literal source joins.